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The Editors, *Journal of Cheminformatics*

Dear Editors,

Please consider the enclosed manuscript, “Local structure–activity landscape roughness detects the activity-cliff failure mode that standard QSAR reliability methods miss,” for publication as a Research article in the *Journal of Cheminformatics*.

QSAR and deep-learning models guide compound prioritization throughout drug discovery, yet they fail on activity cliffs (pairs of near-identical molecules whose potencies differ by orders of magnitude) without warning. The per-compound reliability tools in everyday use do not catch this failure, and our central result makes that concrete with a head-to-head benchmark on the 30-target MoleculeACE data. The four standard per-compound reliability estimators, random-forest ensemble variance, the distance-based applicability domain, Sheridan’s three-measure random-forest error model, and a reliability-density-neighbourhood adaptation, all rank ordinary high-error predictions well but are at chance or worse at identifying activity cliffs (ROC-AUC 0.35 to 0.52), because each is organized around proximity to the training data, whereas cliffs sit in dense, well-sampled regions.

We introduce a per-compound measure of local structure–activity landscape roughness, computed from a compound’s training neighbours without its own activity, that is the only estimator to detect cliffs (median ROC-AUC 0.69), is statistically distinct from the applicability domain under partial-correlation and mixed-effects control, and holds across three model classes, neighbourhood sizes, and distance metrics. Used to condition a conformal predictor, it keeps 90% interval coverage valid on cliffs, where conditioning on ensemble variance or the applicability domain leaves coverage below nominal. Roughness complements rather than replaces uncertainty: a combined score inherits both.

We believe the work fits the *Journal of Cheminformatics* and will interest its readership. It demonstrates a concrete advantage over the established reliability methods rather than proposing a method in isolation; it is model-agnostic and requires no measured activity; and it is entirely reproducible. All data and code are openly available under an OSI-approved open-source licence, with a single `reproduce.sh` entry point, a pinned environment, and a Zenodo-archived snapshot, in keeping with the journal’s reproducibility policy. The relationship also transfers to aqueous solubility and lipophilicity benchmarks.

The manuscript is original, is not published or under consideration elsewhere, and as sole author I declare no competing interests. Thank you for considering this submission.

Sincerely,

Krishna Harish